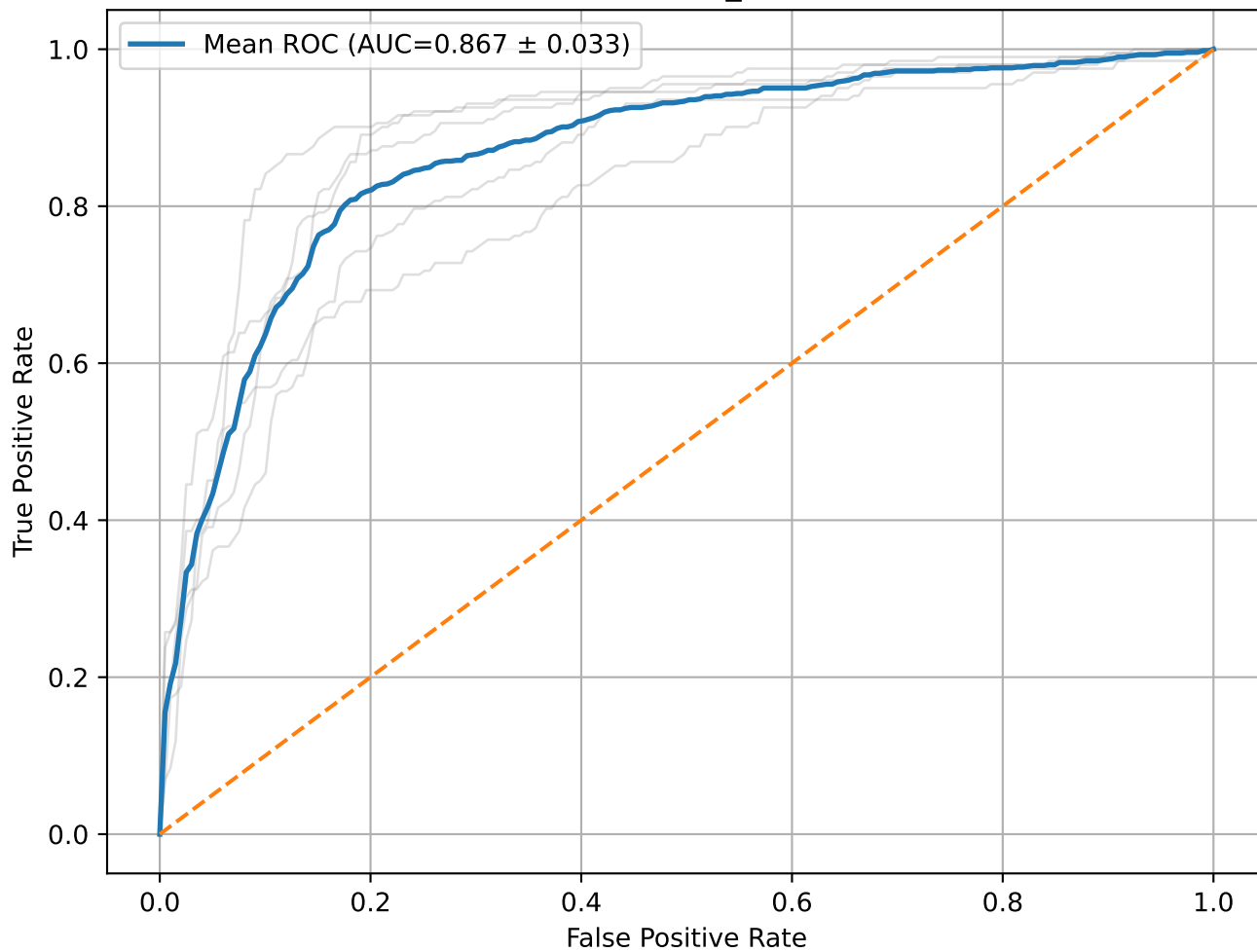


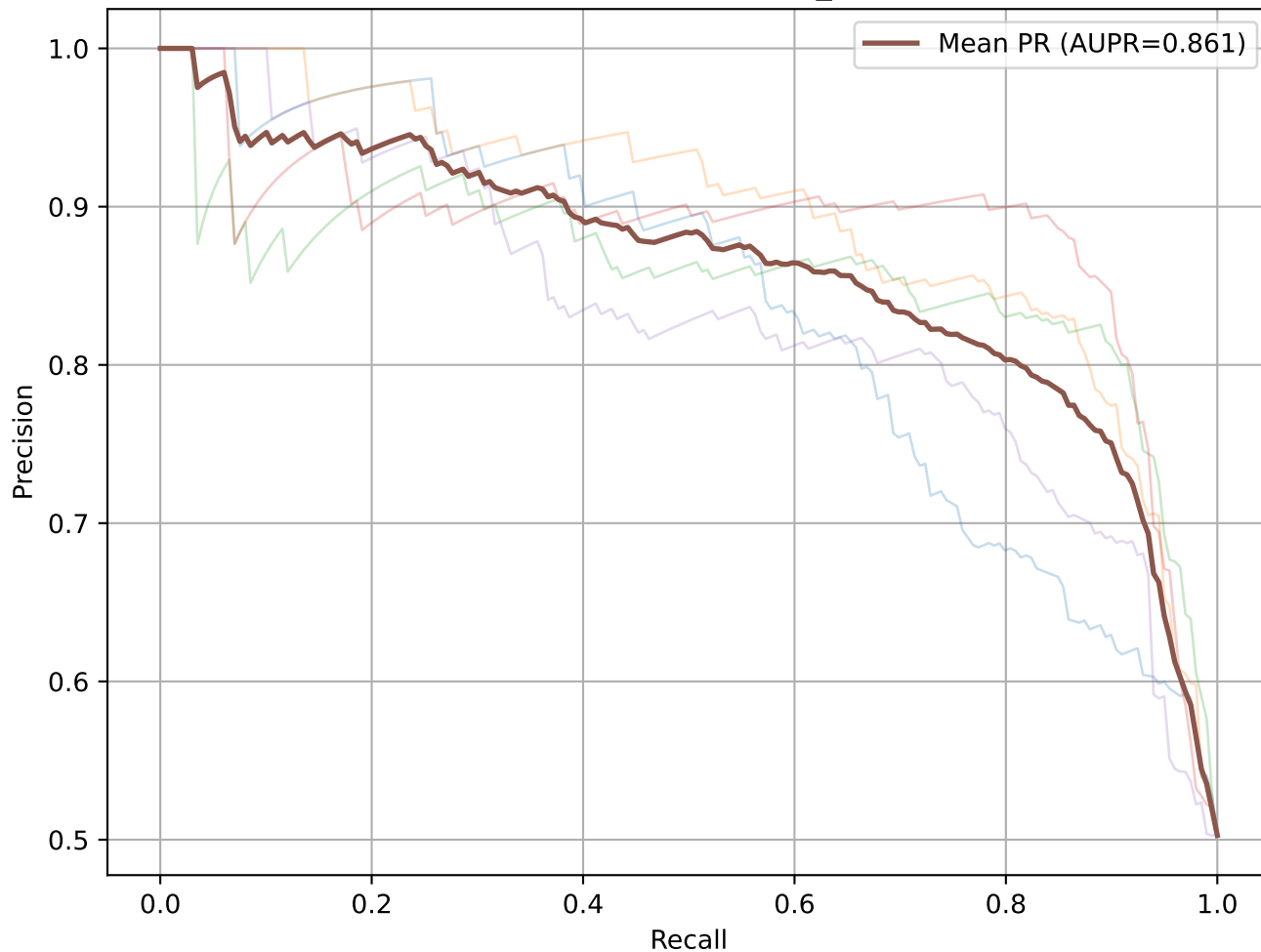
ML Hybrid Models Comparison Report

Generated: 2025-11-23T23:24:54.223084
Dataset samples: 2009, features: 447
Models: pso_svm, bayes_opt_svms, gbdt
StratifiedKFold: 5 splits
Random seed: 42

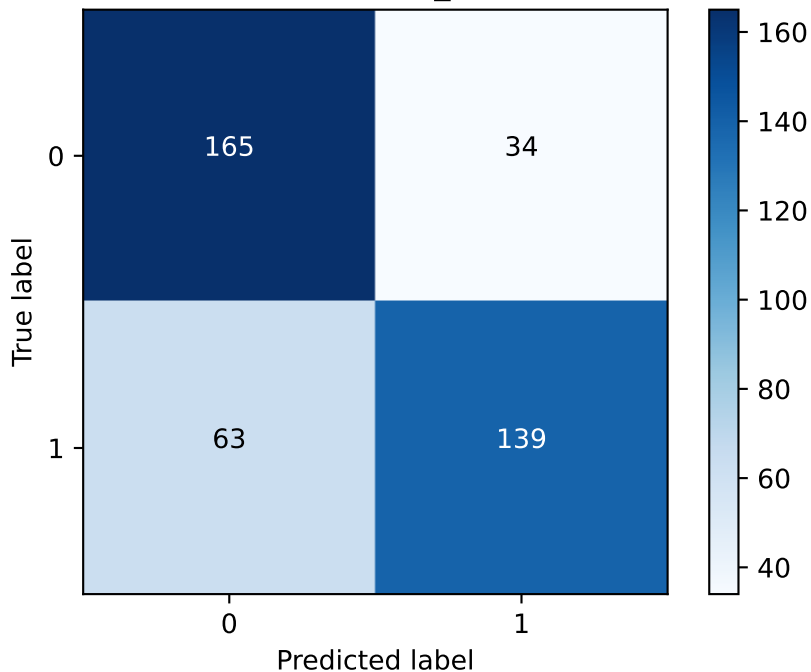
ROC - PSO_SVM



Precision-Recall - PSO_SVM



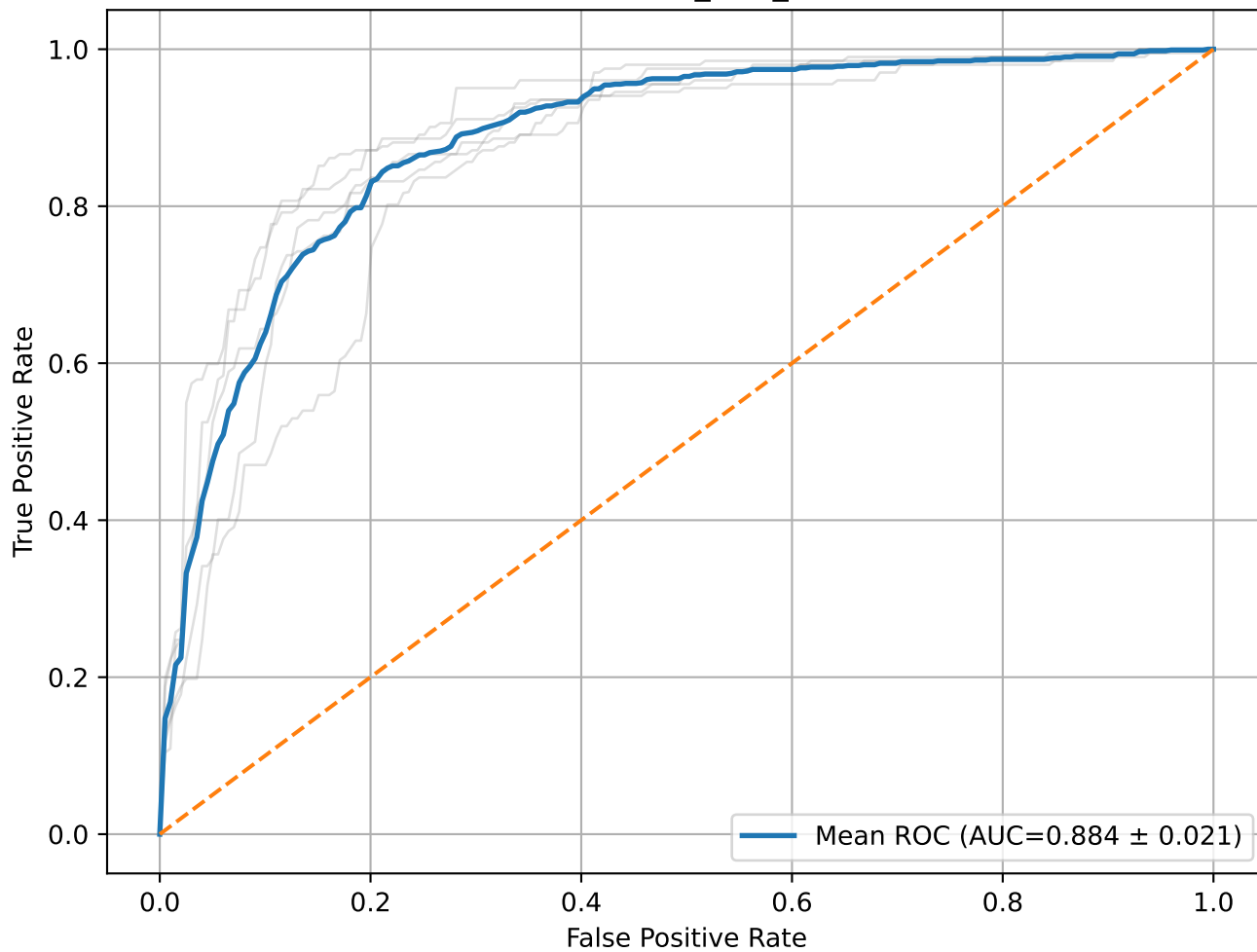
Confusion Matrix - PSO_SVM (last fold)



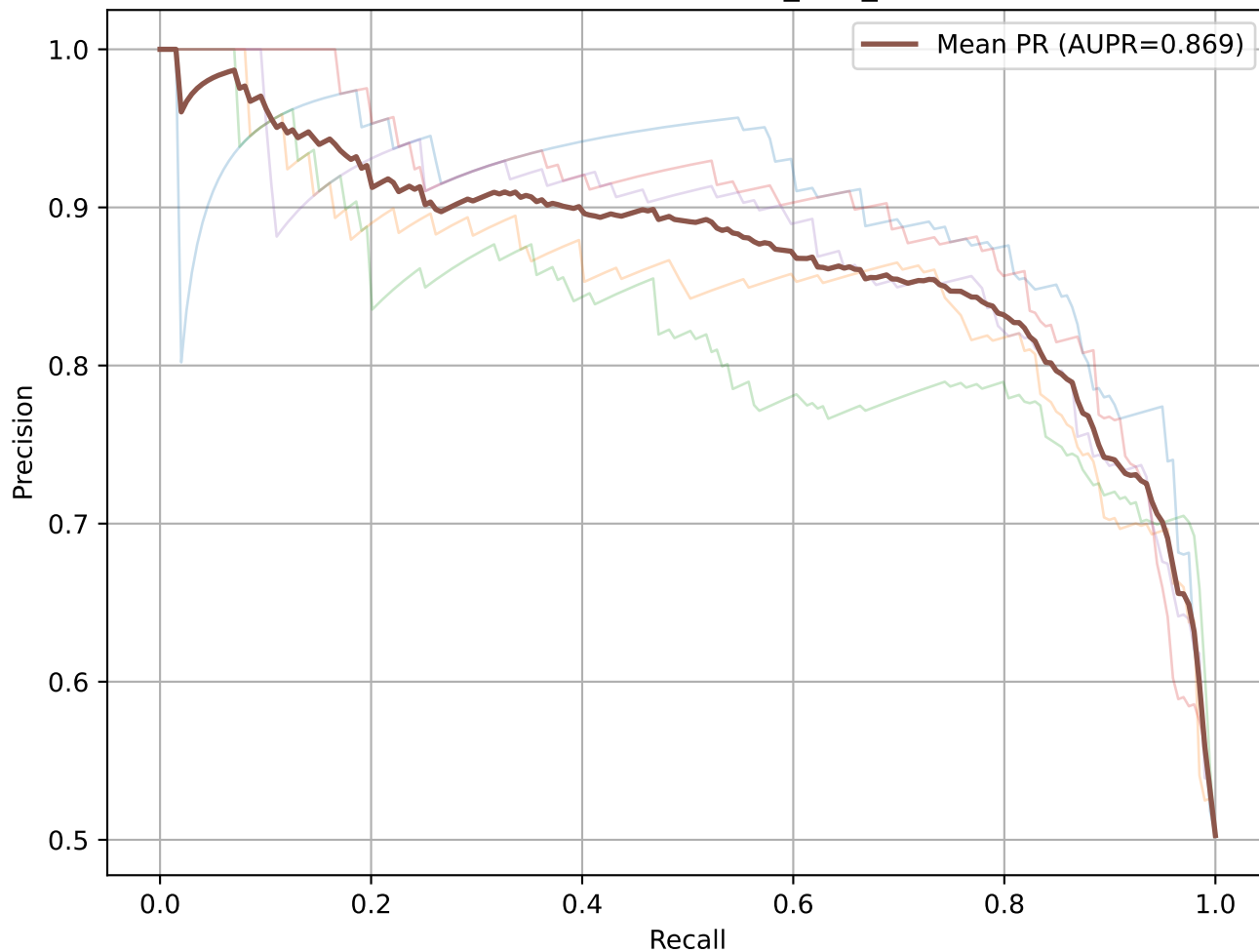
ClassificationReport - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.72	0.83	0.77	199
1	0.80	0.69	0.74	202
accuracy			0.76	401
macro avg	0.76	0.76	0.76	401
weighted avg	0.76	0.76	0.76	401

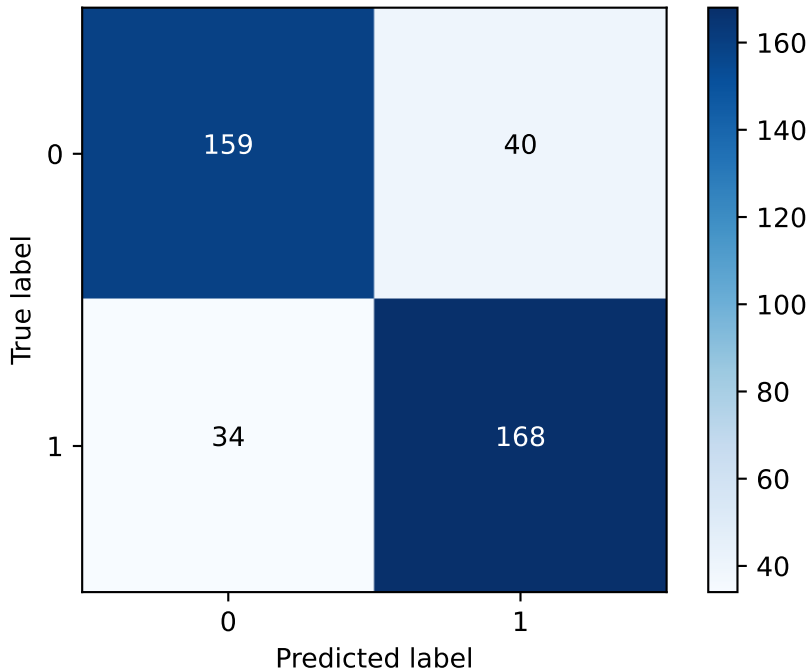
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



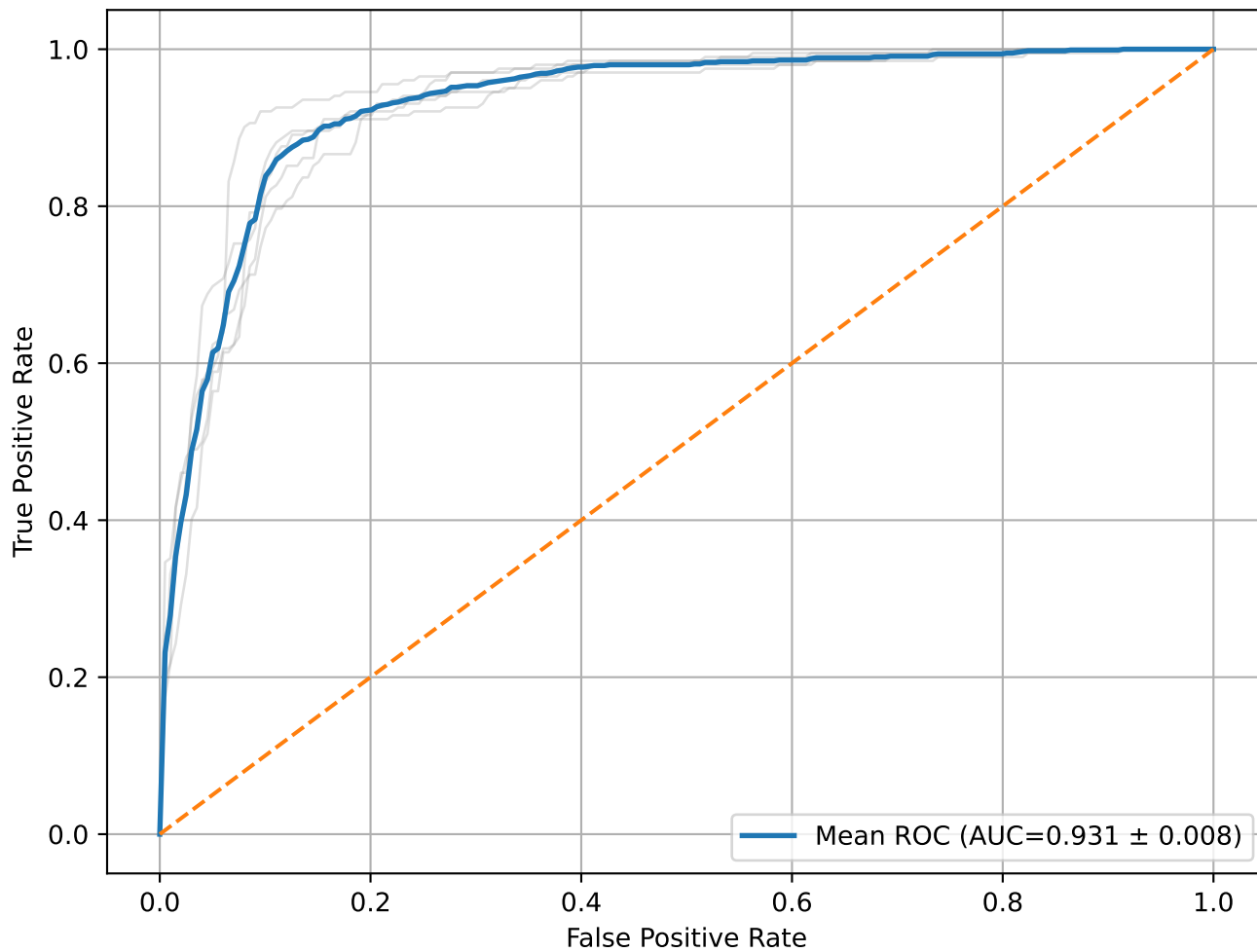
Confusion Matrix - BAYES_OPT_SVMS (last fold)



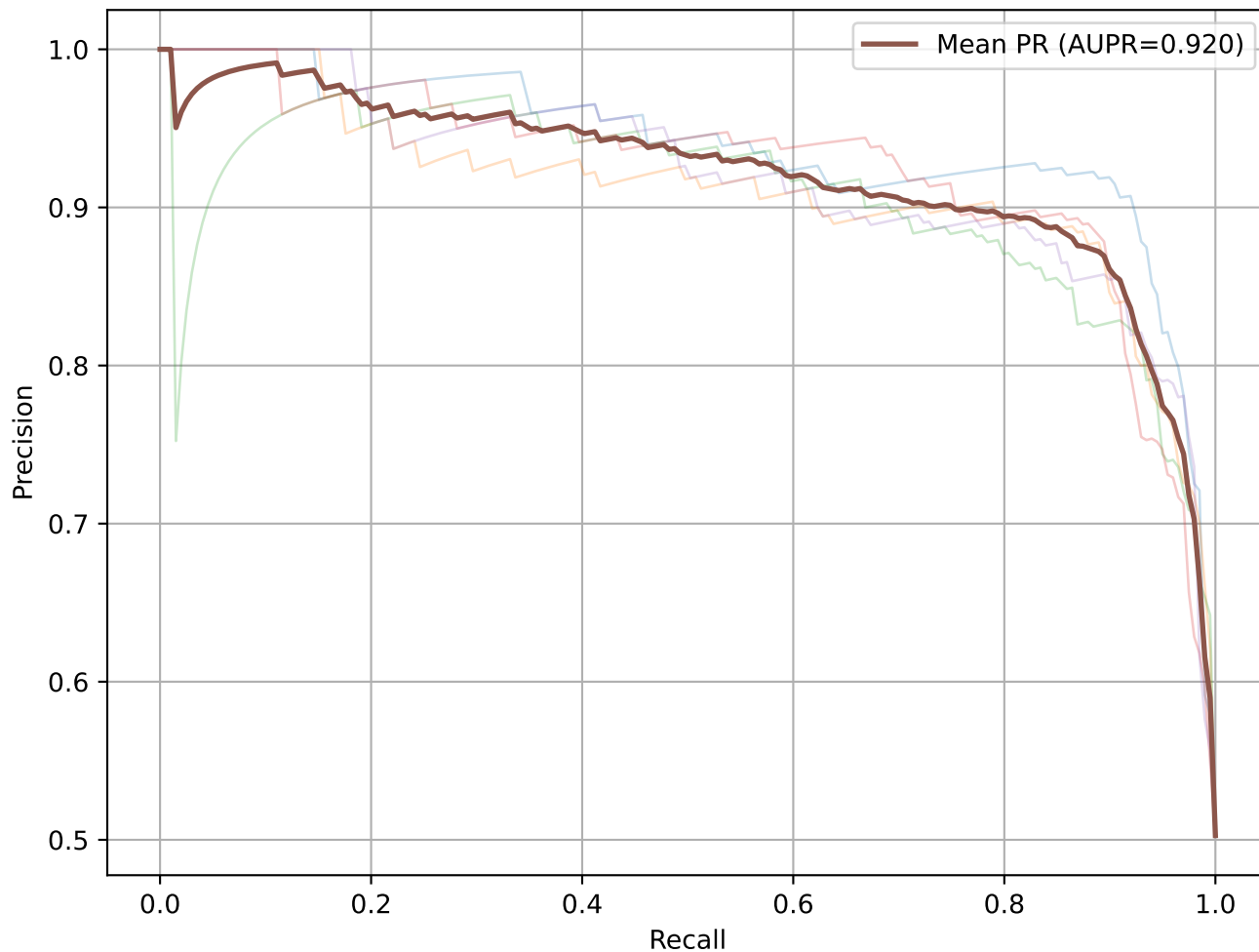
ClassificationReport - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.82	0.80	0.81	199
1	0.81	0.83	0.82	202
accuracy			0.82	401
macro avg	0.82	0.82	0.82	401
weighted avg	0.82	0.82	0.82	401

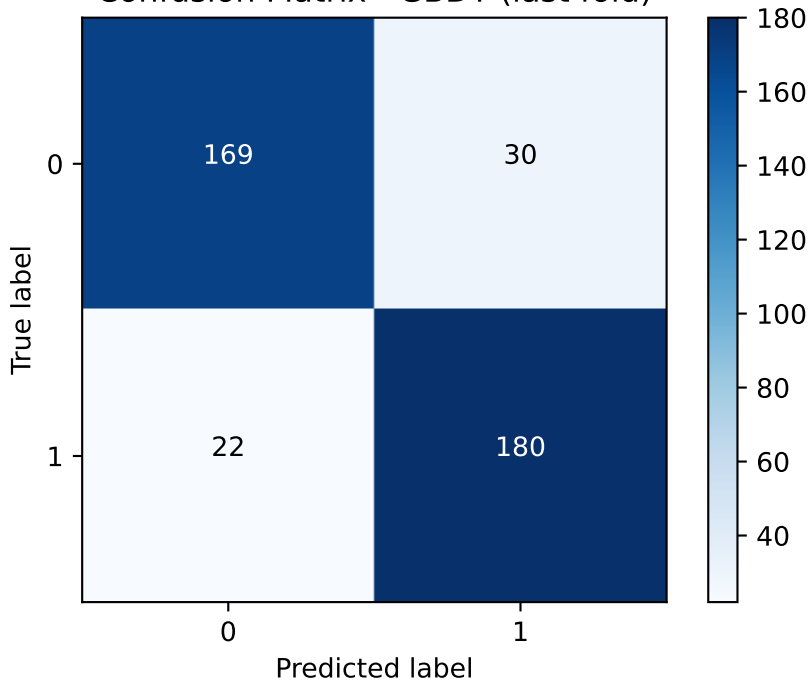
ROC - GBDT



Precision-Recall - GBDT



Confusion Matrix - GBDT (last fold)



ClassificationReport - GBDT (last fold)

	precision	recall	f1-score	support
0	0.88	0.85	0.87	199
1	0.86	0.89	0.87	202
accuracy			0.87	401
macro avg	0.87	0.87	0.87	401
weighted avg	0.87	0.87	0.87	401

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
gbdt	0.929297	0.007901	0.919526	0.876055	0.877476
bayes_opt_svms	0.882009	0.020875	0.868402	0.817819	0.821695
pso_svm	0.865846	0.032650	0.860319	0.809332	0.801165

Combined ROC - All Models

